

Tohma Kuroyanagi

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Education

California State University, Long Beach — Long Beach, CA
B.S. Computer Science | Expected May 2028
GPA: 3.7/4.0

Skills

Languages: C++, JavaScript, Python, SQL, HTML/CSS
Frameworks & Libraries: React, Express.js, Node.js, OpenCV, YOLOv8
Tools & Technologies: Git, Docker, Linux/Unix, Arduino, PySerial, Vercel, Render, VS Code, WordPress, Figma
Concepts: Computer Vision, Embedded Systems, Control Systems, Real-Time Systems, Serial Communication, REST APIs, Full-Stack Development, Object-Oriented Programming, Data Structures & Algorithms
Languages Spoken: English, Japanese

Projects

AI Vision Tracker | Python, OpenCV, YOLOv8, Arduino
Summer 2026

- Developed a real-time AI vision tracking system using YOLOv8 and OpenCV to detect and track people through a servo-actuated camera platform at ~8 FPS.
- Implemented a closed-loop control pipeline integrating computer vision, serial communication, and servo actuation.
- Built Python-to-Arduino communication using PySerial to enable real-time hardware control from tracking signals.
- Tuned tracking performance using smoothing, deadzones, and proportional gain adjustments to reduce oscillation.

BattleBot Control System | Arduino, C++, Embedded Systems
Spring 2026

- Collaborated in a 5-person team to test embedded control systems for a remote-controlled BattleBot.
- Wired ESCs, RC receiver channels, motors, and Arduino Uno for weapon and drivetrain control.
- Tested Arduino programs for RC pulse reading, PWM motor control, ESC arming, and serial debugging.
- Validated weapon motor activation and wheel movement through RC controller input.

AI-Powered Study Assistant | React, Express.js, LLM APIs, Docker
Spring 2026

- Built and deployed a full-stack AI web app that generates summaries, flashcards, and quizzes from study notes.
- Developed a React frontend with markdown rendering, dark mode persistence, loading states, and clipboard support.
- Implemented Express.js REST APIs integrating Groq LLM services for real-time AI-generated educational content.
- Containerized and deployed a 2-service architecture using Docker, Vercel, and Render.

Experience

Marketing Team Intern — Seeknet USA
Summer 2025

- Designed and developed the company's internship landing page using HTML and CSS.
- Created website sitemaps and analyzed client site structures to identify usability improvements.
- Translated Japanese WordPress blog content into English to expand audience reach.
- Assisted with LinkedIn and Instagram marketing campaign planning and content strategy.

Relevant Coursework

Algorithms, Operating Systems, System Programming, Data Structures, Software Engineering, Database Fundamentals